

Evaluating the impacts of carbonaceous aerosols on clouds and climate

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3.1 Introduction

Any attempt to reconcile observed surface temperature changes within the past 150 years to changes simulated by climate models that include various atmospheric forcings is sensitive to the changes attributed to aerosols and aerosol–cloud–climate interactions, which are the main contributors that may well balance the positive forcings associated with greenhouse gases, absorbing aerosols, ozone related changes, etc. These aerosol effects on climate, from various modeling studies discussed in Menon (2004), range from $+0.8$ to -2.4 W/m^2 , with an implied value of -1.0 W/m^2 (range from -0.5 to -4.5 W/m^2) for the aerosol indirect effects. Quantifying the contribution of aerosols and aerosol–cloud interactions remains complicated for several reasons, some of which are related to aerosol distributions and some to the processes used to represent their effects on clouds. Aerosol effects on low-lying marine stratocumulus clouds that cover much of the Earth's surface (about 70%) have been the focus of most prior simulations of aerosol–cloud interaction effects. Since cumulus clouds (shallow and deep convective) are short-lived and cover about 15 to 20% of the Earth's surface, they are not usually considered as radiatively important. However, the large amount of latent heat released from convective towers, and corresponding changes in precipitation, especially in biomass regions owing to convective heating effects (Graf *et al.*, 2004), suggest that these cloud systems, and aerosol effects on them, must be examined more closely. The radiative heating effects for mature deep convective systems can account for 10–30% of maximum latent heating effects and thus cannot be ignored (Jensen and Del Genio, 2003).

The first study that isolated the sensitivity of cumulus clouds to aerosols was from Nöber *et al.* (2003) who found a reduction in precipitation in biomass burning regions and shifts in circulation patterns. Aerosol effects on convection have been included in other models as well (cf. Jacobson, 2002) but the relative impacts on convective and stratiform processes were not separated. Other changes to atmospheric stability and thermodynamical quantities due to aerosol absorption are also known to be important in modifying cloud macro/micro properties. Linkages between convection and boreal biomass burning can also affect the upper troposphere and lower stratosphere, radiation and cloud microphysical properties via transport of tropospheric aerosols to the lower stratosphere during extreme convection (Fromm and Servranckx, 2003). Relevant questions regarding the impact of biomass aerosols on convective cloud properties include the effects of vertical transport of aerosols, spatial and temporal distribution of rainfall, vertical shift in latent heat release, phase shift of precipitation, circulation and their impacts on radiation.

Over land surfaces, a decrease in surface shortwave radiation ($\sim 3\text{--}6 \text{ W/m}^2$ per decade) has been observed between 1960 to 1990, whereas increases of 0.4 K in land temperature that occurred during the same period have resulted in speculations that evaporation and precipitation should also have decreased (Wild *et al.*, 2004). However, precipitation records for the period 1950–2000 over most land areas do not indicate any decrease, or for that matter any significant trend (Beck *et al.*, 2005). Wild *et al.* (2004) speculate that the decrease in precipitation may be related to increased moisture advection from the oceans, which may well have some contributions from

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aerosol–radiation–convection coupling that could modify circulation patterns and hence moisture advection in specific regions.

Other important aspects of aerosol effects, besides the direct, semi-direct, microphysical and thermodynamical impacts, include alteration of surface albedos, especially snow and ice covered surfaces, due to absorbing aerosols. These effects are uncertain (Jacobson, 2004) but may produce as much as 0.3 W/m^2 forcing in the northern hemisphere that could contribute to melting of ice and permafrost and change in the length of the season, such as the early arrival of spring (Hansen and Nazarenko, 2004). Besides the impacts of aerosols on the surface albedos in the polar regions, and the thermodynamical impacts of Arctic haze (composed of water-soluble sulfates, nitrates, organic and black carbon (BC)), the dynamical response to Arctic haze (through the radiation–circulation feedbacks that cause changes in pressure patterns) is thought to have the potential to modify the mode and strength of large-scale teleconnection patterns such as the Barents Sea Oscillation that could affect other climate regimes, mainly in Europe (Rinke *et al.*, 2004). Additionally, via the Asian monsoon, wind patterns over the eastern Mediterranean and lower stratospheric pollution at higher latitudes (Lelieveld *et al.*, 2002) are thought to be linked to the pollutants found in Asia, indicating the distant climate impacts of aerosols.

Thus, it becomes difficult to quantify regional changes in precipitation or radiation in the context of a regional model, since long-range transport of aerosols and its dynamical impacts can modify distant climates and can affect boundary conditions that are imposed in regional simulations. On the other hand, the use of a global model to investigate regional changes may lead to problems associated with not resolving processes on coarse spatial scales. Here, we will examine global model simulations of the indirect effect for warm clouds, both stratiform and convective, in order to examine the influence of aerosols in modifying precipitation and radiation. Since stratus clouds cover large areas, spatial domain is not a constraint, whereas for convective clouds in biomass regions, the area covered by biomass burning may be large enough to examine regional influences. Section 3.2 describes the model and simulations that were performed in this study, Section 3.3 describes the results from the modeling study as well as results obtained for different regions, and finally in Section 3.4 we present the conclusions from this study.

3.2 Model description

We use the Goddard Institute for Space Studies (GISS) ModelE, the latest version of the GISS model II', which is described in more detail by Schmidt *et al.* (2005) and Hansen *et al.* (2005), coupled to the aerosol chemistry model of Koch *et al.* (1999, 2005). Cloud water is treated prognostically in the model (Del Genio *et al.*, 1996), with representation of

sources due to large-scale convergence and cumulus detrainment, and sinks due to autoconversion, accretion, evaporation, and cloud top entrainment, as well as precipitation enhancement due to the seeder-feeder effect. Stratiform cloud cover is a diagnostic function of relative humidity and stability allowing for subgrid vertical cloud fraction. Moist convection uses a quasi-equilibrium cloud base closure with entraining and non-entraining updrafts and a cumulus-scale downdraft. Main changes in the updated GCM of relevance to the indirect effect include a microphysics-based cumulus scheme described in Del Genio *et al.* (2005), and subgrid turbulence at all levels of the GCM (based on the second-order closure model of Cheng *et al.* 2003). Improvements to the cumulus scheme include a more physical partitioning of convective condensate between precipitation and detrainment into anvil clouds based on a microphysics scheme (using a Marshall–Palmer distribution for droplets, empirical relationships between droplet size and terminal velocity for liquid, graupel, and ice hydrometeors, and prescribed cumulus updraft speeds).

The aerosol emission data sets used in the model are derived from AEROCOM (an aerosol model intercomparison project; Frank Dentener, personal communication, 2004). Sulfate aerosols include fossil fuel sources that are country-based emissions for the year 2000 and both continuously erupting and explosive volcanoes (Andres and Kasgnoc, 1998; Halmer and Schmincke, 2003) as well as dimethyl sulfide (DMS) emissions. Carbonaceous aerosols (treatment described in Koch and Hansen, 2005) include fossil- and biofuel sources from Bond *et al.* (2004) and biomass sources. Biomass sources for sulfates and carbonaceous aerosols are from van der Werf *et al.* (2004). In addition, organic aerosols include a fraction of natural terpenes (15%) from secondary organics. Solubility for industrial carbonaceous aerosols is 100% for aged aerosols (e-folding time of 1 day). For the biomass components, solubilities of 80% and 60% are assumed for organic and BC, respectively. Sea-salt is also calculated based on model wind speeds (Koch *et al.*, 2005) instead of through the source function used previously (Koch *et al.*, 1999). All aerosols are treated as external mixtures, which may not accurately represent aerosol radiative properties for regions away from sources, where aerosols tend to form internal mixtures.

Improvements to the treatment of the aerosol indirect effect described in Menon *et al.* (2002a) (hereafter referred to as M02) include a semi-prognostic treatment of the cloud droplet number concentration (CDNC), addition of BC aerosol effects on clouds, and incorporation of subgrid vertical velocity effects on cloud droplet number. Several parameterizations have been used to determine CDNC, which is the main and critical link between the aerosols and cloud microphysics. The resulting values for the indirect effect vary by $\sim 40\%$, depending on the assumptions or parameterizations used. Kiehl *et al.* (2000) find over a factor of 3 difference in their estimate of the indirect effect when using different CDNC

schemes. Here, we mainly refer to simulations that use the parameterization from Gultepe and Isaac (1999), given as:

$$\begin{aligned} \text{CDNC}_{\text{Land}} &= 298 \times \log_{10} N_{a, \text{land}} - 595 \\ \text{CDNC}_{\text{Ocean}} &= 162 \times \log_{10} N_{a, \text{ocean}} - 273 \end{aligned} \tag{3.1}$$

where N_a is the aerosol concentration (cm^{-3}) calculated from the aerosol mass concentration and assumed sizes as used in Lohmann *et al.* (1999). The aerosol concentration N_a for land includes the effects of sulfates, organic and BC aerosols, whereas N_a for ocean includes, in addition, sea-salt aerosols. The values so derived for CDNC are representative of in-cloud CDNC values since measurements were for in-cloud values. Dust is assumed to be insoluble in these simulations and since the amount of dust roughly remains unchanged for present-day and pre-industrial simulations, its contribution to the indirect effect in terms of changes in CDNC is neglected. Future treatments will include dust formed on sulfate or sea-salt via heterogeneous chemical reactions (Bauer *et al.*, 2005), which is not discussed in this work. Aerosol sources for present-day simulations include both anthropogenic (sulfates and carbonaceous aerosols from fossil fuel, biofuel and biomass sources) and natural sources from DMS, volcanic and biomass emissions (assumed to be 50% of the anthropogenic biomass source), organic aerosols from terpenes, sea-salt and dust. Aerosol sources for pre-industrial simulations include just the natural sources. Where applicable, we use “ Δ ” to denote differences between present-day and pre-industrial simulations. All simulations use fixed sea-surface temperatures derived for present-day climatologies. All results reported are based on 6 years of model simulations averaged over the past 5 years.

3.3 Aerosol indirect effect on warm clouds

Prior model estimates of the aerosol indirect effect (AIE) with the GISS GCM (M02) are based on simulations that distinguish between pre-industrial and present-day aerosol emissions. These estimates depend on the background concentration imposed for either aerosols or CDNC. As an example, in climate simulations for present-day versus pre-industrial aerosols, changing the minimum (background) CDNC from 10 cm^{-3} to 40 cm^{-3} produces almost a 50% change in the indirect effect estimate (Menon, 2004). Here, we use a value of 20 cm^{-3} for background CDNC for all our simulations based on surface observations that indicate values of ~ 10 to 20 cm^{-3} in pure marine background air with no wind (J.-L. Brenguier, personal communication). We perform several sets of simulations, shown in Table 3.1, to investigate the impacts of fossil- and biofuel BC aerosols, aerosol-convective cloud effects, and the semi-direct aerosol effects. Note that all simulations only include aerosol effects on warm cloud microphysics and that they include both the first indirect effect (changes in cloud reflectivity due to smaller droplet sizes;

Table 3.1 Model simulations and their description.

Simulation	Type
Exp A	Standard run with both types of aerosol indirect effects
Exp A_S	Like Exp A but with changes to the aerosol burden ($2 \times$ anthropogenic sulfate for present-day aerosol burden and 10% of present-day biomass aerosols for pre-industrial aerosol burden)
Exp NBC	Like Exp A but without any fossil- and biofuel sources for BC
Exp 2BC	Like Exp A but with twice the fossil- and biofuel sources for BC
Exp CC1	Like Exp A but including aerosol effects on convective clouds
Exp CC2	Like Exp CC1 but with a different treatment for determining cloud droplet number concentration in convective clouds
Exp NIE	Like Exp A but with fixed cloud droplet number

Twomey, 1977) and the second indirect effect (changes in cloud lifetime and cloud cover from suppression of precipitation; Albrecht, 1989). Our indirect effect represents the net effects of aerosols on clouds since it includes implicit changes to cloud fields from other climate feedbacks.

Additional simulations similar to those listed in Table 3.1 were performed that include aerosol emissions for pre-industrial scenarios (year 1850). Present-day emissions (year 2000) include both natural and anthropogenic aerosols. Most of the treatment for the aerosol indirect effect follows that given in M02, except that we modulate the CDNC so that it changes with cloud cover and cloud water changes. Here, we use the formula of Beheng (1994) for autoconversion (the process that initiates the conversion of cloud water to precipitation), which is a function of cloud liquid water content and CDNC. The magnitude of the indirect effect, calculated as difference between present-day and pre-industrial simulations, for Exp A is about -1.36 W/m^2 obtained from differences between net radiation at the top of the atmosphere (TOA) and the direct aerosol effect.

Alternatively, the aerosol indirect effect can be obtained from the difference in net cloud radiative forcing, which then is -0.65 W/m^2 . Differences between these methods of estimating the indirect effect, which are both approximate, are discussed in M02. They are dependent on the treatment used to separate radiation into that due to aerosols or that due to clouds. These values are significantly lower than our previous estimates which ranged from -1.55 to -4.4 W/m^2 for the indirect effect (change in net cloud forcing) and could be attributed to the minimum CDNC used here (as opposed to

Table 3.2 Globally averaged annual values of aerosol column burdens (mg/m²) for the different simulations.

Two values listed for M02 are for two sets of simulations of the first and second indirect effects that mainly differ in the treatment of the autoconversion scheme. The values separated by '/' are for present-day and pre-industrial aerosol burdens. Note that in M02 only total organic aerosol burden was available, which includes the biomass/terpene components.

Case	Sulfate total	OC (fossil- & biofuel)	OC (biomass & terpene)	BC (fossil- & biofuel)	BC (biomass)	Net cloud forcing (W/m ²)
M02	2.66/0.42	1.57/0.14		–	–	–4.36
	5.03/1.05	2.46/0.27		–	–	–2.41
Exp A	2.96/0.15	0.98/0.57	1.61/0.80	0.13/0.0	0.12/0.06	–0.65
Exp A_S	4.34/0.14	0.96/0.55	1.63/0.15	0.12/0.0	0.12/0.01	–1.03

10 cm^{–3} in M02); inclusion of dust and BC aerosols (which will increase the positive forcing); inclusion of a relative humidity factor that accounts for changes in extinction efficiencies due to aerosol growth (~36% increase in net cloud forcing); choice of autoconversion parameterization and other general improvements in Model E (e.g. improvements to the convective scheme resulting in increased coverage of marine stratus clouds along the western edges of the continent, which were previously deficient; Del Genio *et al.*, 2005). Perhaps the more significant reason for the smaller values obtained here is the smaller anthropogenic aerosol burden and the larger assumed natural aerosol burden (increased biomass aerosols assumed for pre-industrial conditions compared with Koch *et al.*, 1999) as shown in Table 3.2. In Exp A, we take anthropogenic aerosol column burdens to be 2.3 times the assumed natural aerosol burden; in M02 they were 4.6 to 6.5 times the natural burden. This smaller burden leads to the large reduction in the aerosol indirect effect magnitude.

As a sensitivity test, we performed an additional simulation, similar to Exp A but with increased anthropogenic sulfates (factor of 2 increase) for present-day conditions and one with reduced biomass aerosols (10% of present-day) for pre-industrial conditions (Exp A_S). The resulting anthropogenic column burden in Exp A_S is higher by a factor of 7 compared with the assumed natural burden. Compared with ΔExp A, the magnitude of the indirect effect increases by 60% for ΔExp A_S, mainly owing to the higher anthropogenic burden in Exp A_S. As shown in Table 3.2, in addition to the choice of the autoconversion parameterization (which leads to some of the differences in the two M02 simulations), aerosol burdens play a major role in determining changes in radiative forcing.

3.3.1 Black carbon aerosol effects on clouds

As shown in Table 3.3, the forcing efficiency, defined as the ratio of direct forcing to anthropogenic column burden, of fossil- and biofuel BC is quite high. Since the heating associated with these absorbing aerosols can lead to significant changes in surface energy budgets, precipitation, cloud cover and circulation (Ramanathan *et al.*, 2001; Chung *et al.*, 2002;

Table 3.3 Values of direct forcing and forcing efficiencies for the different species for Exp A.

Case	Sulfate total	OC (fossil- and bio-fuel, biomass)	BC (fossil- and biofuel)	BC (biomass)	Total
Direct forcing (W/m ²)	–0.29	–0.13	0.18	0.06	–0.18
Forcing efficiency (W/g)	–103	–106	1385	857	NA

Menon *et al.*, 2002b), and since the sources for fossil- and biofuel BC may be controlled more readily (via controls on technology, e.g. emission filters for transportation, clean-coal burning technology, use of natural gas to replace biofuels, etc.) than that for biomass-derived aerosols, a closer inspection is warranted to understand the impacts of these absorbing aerosols on clouds and climate.

Present-day climate simulations (with twice the BC from fossil- and biofuel sources (Exp 2BC); with the standard amount (Exp A); and without any fossil- and biofuel BC (Exp NBC)) were compared with climate simulations for pre-industrial aerosol concentrations. For annual global changes, the major differences made by increasing amounts of BC were decreased net radiation (solar) at the surface, and decreased sensible heat flux, precipitation and soil evaporation. Increases in net TOA radiation, shortwave cloud forcing and water cloud optical depth were also observed for increasing BC. With respect to simulations for pre-industrial aerosol concentrations, these changes were within 2.5% except for changes to water cloud optical depths (>10%), net TOA radiation (>15%) and net heating (>15%) at the surface. Differences of 3% are also manifested in cloud top temperature (which decreases with increasing BC) for varying BC amounts. While global differences are not as high, regional changes are more significant and are discussed in more detail in Section 3.3.3. Although changes in optical depth are related to changes in

cloud water paths and cloud droplet sizes, the increase in cloud optical depth is not related to changes in CDNC ($\sim 5\%$ between Exp A, 2BC and NBC), but rather to changes due to the thermodynamical effects of heating (sensible and latent heat fluxes) that change the availability of water since precipitation is reduced.

Also of significance are changes to the zonally averaged vertical profiles of temperature at higher latitudes, which increase with increasing BC especially towards the poles and higher levels (indicative of upper tropospheric transport of warm air), and solar radiation heating rates that are much stronger in the northern hemisphere. This is not surprising since the impacts of BC-related forcing tend to be stronger in regions of high surface albedo (Hansen and Nazarenko, 2004). Increasing amounts of BC also lead to greater reduction in the extent of snow and ice cover (reductions seen in $\Delta\text{Exp 2BC}$ and $\Delta\text{Exp A}$ are a factor of 1.6 higher than those in $\Delta\text{Exp NBC}$), as was also indicated by Hansen and Nazarenko (2004), Wang (2004) and Roberts and Jones (2004); with stronger changes in the northern hemisphere and the Arctic than in the Antarctic. Also of significance is the change in sea-level pressure over all ocean regions and part of the NH continents (positive anomalies in $\Delta\text{Exp 2BC}$ replace neutral changes found in $\Delta\text{Exp A}$ and $\Delta\text{Exp NBC}$) indicative of changes in the circulation patterns that are stronger in $\Delta\text{Exp 2BC}$. Similar to the results of Wang (2004), we find changes in circulation and snow cover, as well as surface fluxes, with changing BC amounts. Analyses of the June to August precipitation fields for Exp A, NBC and 2BC do indicate a northward shift in the Intertropical Convergence Zone (ITCZ) in Exp 2BC and Exp A, as was also observed in simulations of BC climate effects by Wang (2004) and Roberts and Jones (2004). Without fossil- and biofuel BC this northward shift in the ITCZ is not evident.

Other uncertainties, besides vertical and horizontal distributions, are related to absorption properties of BC within clouds. In-cloud absorption by BC is thought to increase the absorption efficiency of clouds by 5% globally, with more significant regional changes (15–25%; Chuang *et al.*, 2002). These results were for the first indirect effect only. Accounting for these BC absorption effects in cloud for both indirect effects, via a proxy enhancement factor (25%) that is analogous to increasing BC optical depth, results in a 12% increase in net cloud forcing (sum of shortwave and longwave) globally. Regionally, differences are much greater in regions with large BC emissions and co-located clouds. In terms of surface temperature response with BC in-cloud heating included, as shown in Figure 3.1 (see color plate section), global differences are negligible but regional differences are much greater in regions where BC emissions dominate, such as Asia, Europe and eastern United States. (Figure 3.2, in the colour plate section, shows the annual column burdens of carbonaceous aerosols from biomass and fossil- and biofuel sources.) Note that these responses are mainly a sensitivity test to indicate the lower bound of the change in temperature when BC in-cloud

absorption is accounted for, since we use an atmosphere-only model with prescribed sea surface temperatures (Wang, 2004). A coupled ocean–atmosphere model would be more effective in simulating the full impact of observed surface temperature changes.

In an atmosphere model (Hadley Centre climate model) coupled to a slab-ocean model with four times as much fossil fuel BC as in Exp A, annual mean surface temperature change is $\sim 0.436\text{ K}$ giving a climate sensitivity of $0.56\text{ K}/(\text{W}/\text{m}^2)$ (Roberts and Jones, 2004). However, effects of BC on cloud properties are not explicitly parameterized in these simulations. Within the same model the climate sensitivity to doubled CO_2 is about $0.91\text{ K}/(\text{W}/\text{m}^2)$. Another study on temperature response per unit of direct forcing (Jacobson, 2002), without the use of a slab-ocean model, reports a value of $1.4\text{ K}/(\text{W}/\text{m}^2)$ for BC and OC from fossil fuel. While a direct comparison to these values is not possible in the context of our simulations (since they include different methods on treatment of carbon optics and climate effects), we can compare the ratios of the relative magnitude of surface temperature changes to radiative flux values between $\Delta\text{Exp A}$, $\Delta\text{Exp NBC}$, and $\Delta\text{Exp 2BC}$. These simulations mainly differ in terms of the amounts of fossil- and biofuel BC. The values for $\Delta\text{Exp A}$, $\Delta\text{Exp NBC}$, and $\Delta\text{Exp 2BC}$ are 0.12, 0.097, and $1.14\text{ K}/(\text{W}/\text{m}^2)$, respectively. The sensitivity of this version of the model coupled to a mixed ocean slab model to doubled CO_2 is $\sim 0.66\text{ K}/(\text{W}/\text{m}^2)$ (Hansen *et al.*, 2005). Although we cannot directly compare these BC and CO_2 -related climate changes from the GISS GCM, because a coupled ocean–atmosphere model must be used to accurately estimate the climate sensitivity to BC, results from Roberts and Jones (2004) and Hansen *et al.* (2005) do suggest that climate sensitivity factors for BC may be quite comparable to that for CO_2 .

3.3.2 Aerosol effects on convective clouds

Besides the radiative effects of fossil- and biofuel aerosols, aerosol effects from biomass burning are also important, especially near regions that are more active convectively. Here we examine aerosol effects on convective clouds, in addition to stratiform clouds. Since parameterizations for predicting CDNC from aerosols for convective clouds are not as well developed as those for stratiform clouds, we use the same droplet prediction scheme as for stratiform clouds (Exp CC1), following the approach of Nuber *et al.* (2003). Using empirical data from Lahav and Rosenfeld (2000), we modify the precipitation rate as a function of cloud base CDNC and temperature, such that for temperatures below 263 K and for low CDNC values ($< 750\text{ cm}^{-3}$), precipitation remains unchanged; and with increasing CDNC and temperatures above 263 K precipitation reduces to 25% ($750\text{ cm}^{-3} < \text{CDNC} < 1000\text{ cm}^{-3}$) and 0% ($\text{CDNC} > 1000\text{ cm}^{-3}$), respectively. These simulations only account for aerosol effects on warm convective clouds.

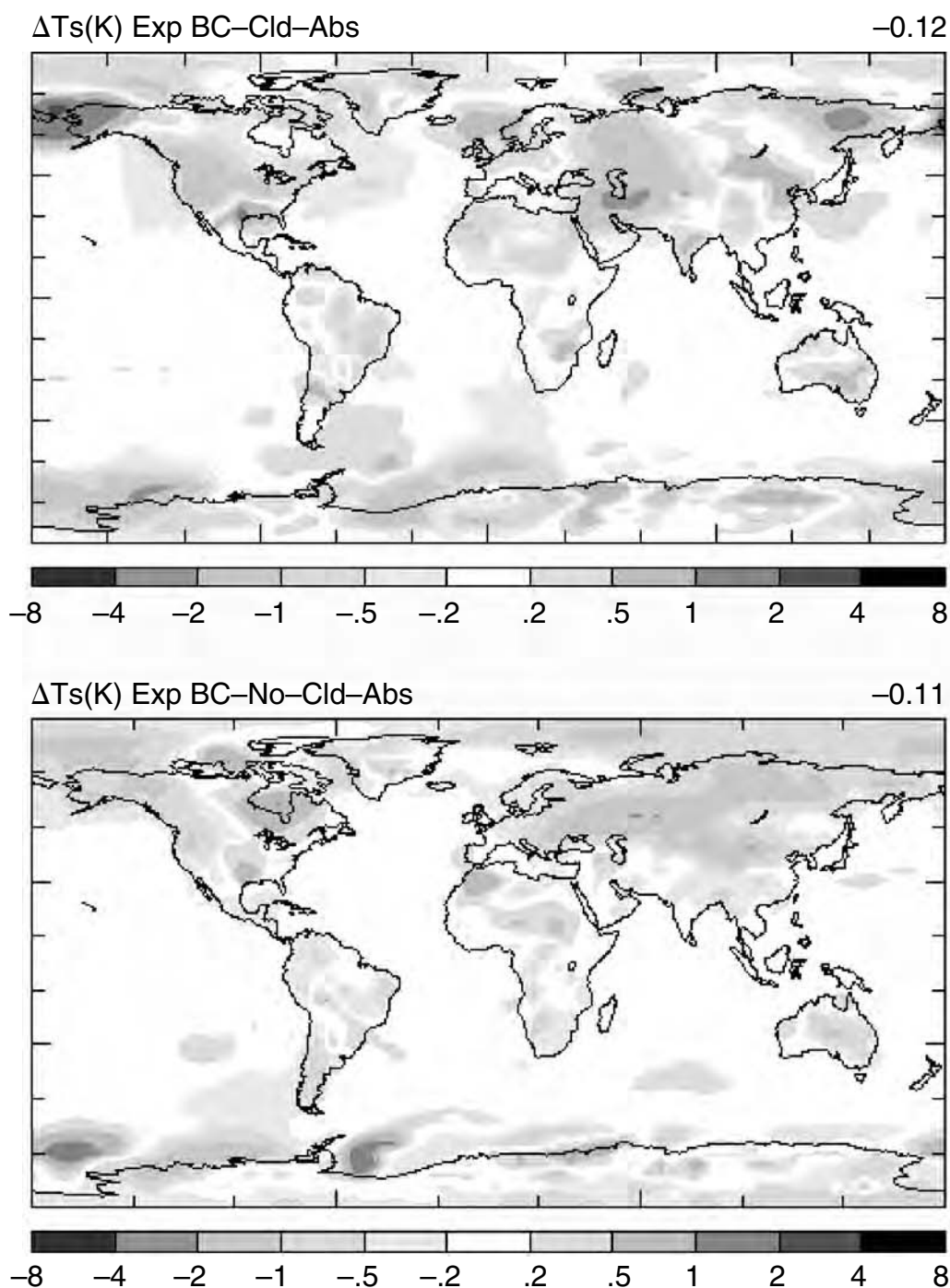


Figure 3.1 Model simulated annual surface temperature change (K) for year 2000 – Year 1850 for simulations that account for BC absorption in-cloud (top panel) and that do not account for BC (bottom panel).

The suppression of rain in warm convective clouds should lead to increased water contents and therefore should also have an effect on freezing rates for cold clouds and precipitation changes (Andreae *et al.*, 2004) via the Bergeron–Findeisen

process, which facilitates ice-phase precipitation processes at the expense of warm-phase processes.

In addition to Exp CC1, we conducted a second simulation (Exp CC2) with CDNC based on observations from the Cirrus

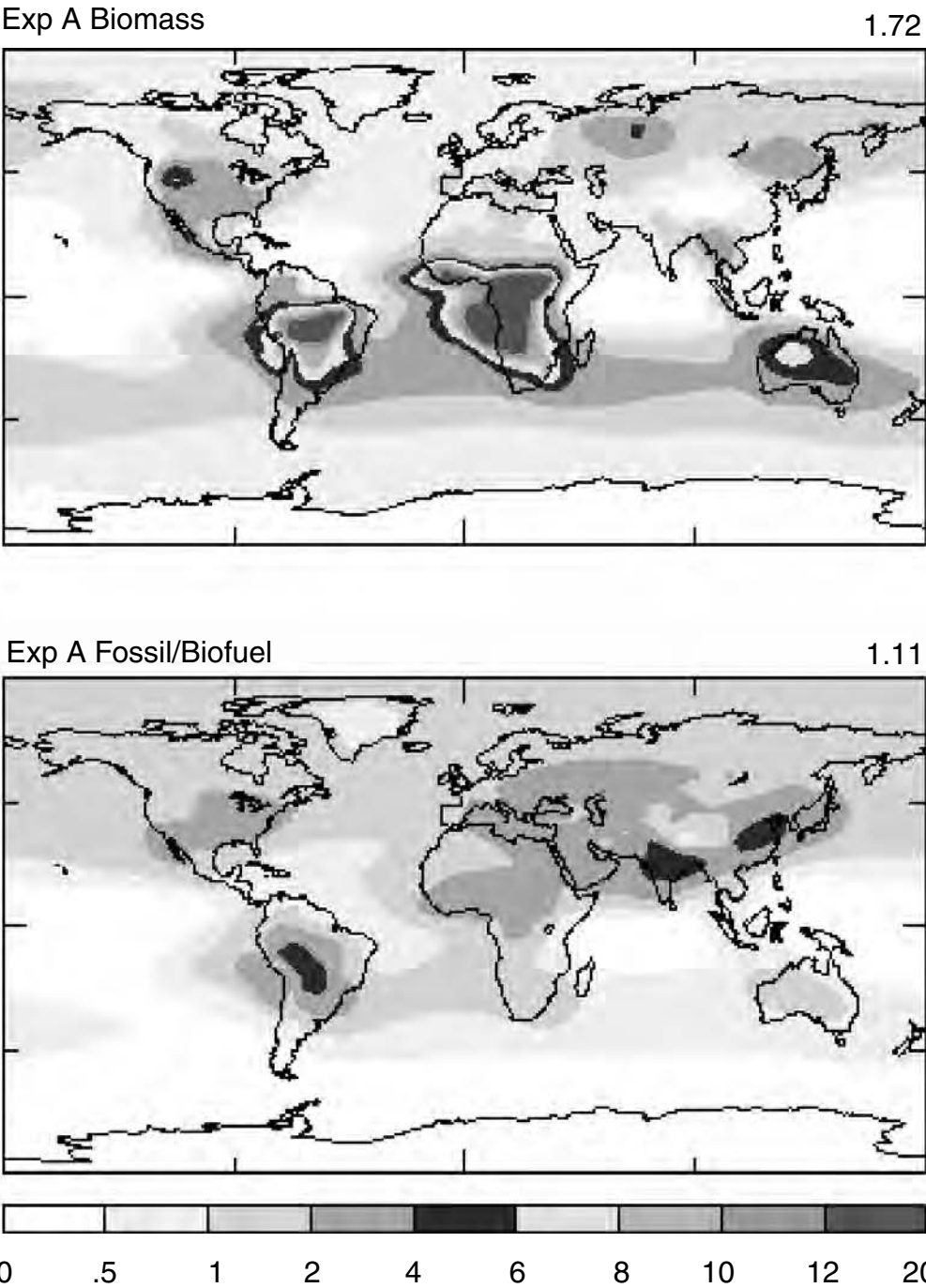


Figure 3.2 Annual values of carbonaceous aerosol column burden distribution (mg/m^2) from biomass (top panel) and fossil- and biofuel sources (bottom panel). Global mean values are on the right-hand side of the figure.

Regional Study of Tropical Anvils – Florida Area Cirrus Experiment (CRYSTAL FACE) data for warm cumulus clouds (Conant *et al.*, 2004). The relationship we use is given as:

$$\text{CDNC} = 10^{[0.433 + 0.815 \log N_a + 0.280 \log(w)]} \tag{3.2}$$

Here, w is the updraft velocity in m/s. Since model diagnostics do not account for cloud-scale updraft velocities we use the following notation

$$w = \bar{w} + \sqrt{0.66 \times \text{TKE}} \tag{3.3}$$

where $\bar{w}$ is the grid average updraft velocity and TKE is the turbulent kinetic energy (Ye Cheng, personal communication). Most of the climate diagnostics for $\Delta\text{Exp CC2}$ are similar to those for $\Delta\text{Exp CC1}$ except for larger differences in the net radiation at TOA, cloud water optical depths, and net heating at the surface, leading to differences in soil evaporation, sensible heat flux, and precipitation. Larger differences are seen regionally and are discussed in Section 3.3.3.

Changes with the addition of aerosol-convective cloud effects (Exp CC1/CC2 versus Exp A) include increases in the liquid water path (40%), water and ice cloud optical depths (increases 20%, and over a factor of 2, respectively), net TOA radiation and net ground heating (70% decrease), shortwave (16%) and longwave (7%) cloud forcing. Increase in cloud cover (water or ice, convective or total) is less than 5% and reduction in total precipitation is between 2 and 10%. With the increase in cloud water (almost a factor of 2), effective water cloud particle sizes (product of cloud effective radii and optical depth) are $\sim 50\%$ larger. More importantly, with aerosol-convective cloud effects the level of precipitation formation may be shifted higher since most of the cloud water increase is at higher levels. Shifts from warm to cold precipitation processes should increase the height at which latent heat is released in clouds with similar rainfall amounts as well as increasing the water available for ice processes, lightning and more intense convective storms (Andreae *et al.*, 2004).

Changes in aerosol concentration for present-day versus pre-industrial simulations (higher by a factor of 2.7) result in values of 0 and 0.22 W/m^2 for the indirect effect from changes in net cloud radiative forcing for $\Delta\text{Exp CC1}$ and $\Delta\text{Exp CC2}$, respectively, and -0.43 W/m^2 for both $\Delta\text{Exp CC1}$ and $\Delta\text{Exp CC2}$ when accounting for the difference in net TOA radiation and the direct effect. Despite similar increases in aerosol burdens, between $\Delta\text{Exp CC1}$, $\Delta\text{Exp CC2}$, and $\Delta\text{Exp A}$, this reduction in the net cloud radiative forcing obtained when including aerosol-convective cloud effects may arise from changes in the longwave component that are somewhat comparable to the shortwave. Some differences may also arise from changes in aerosol burden as indicated from a sensitivity test conducted to understand changes in climate diagnostics for differences in biomass aerosols. At present, we assume that pre-industrial biomass aerosols are $\sim 50\%$ of present-day biomass emissions. These estimates are highly uncertain because of the uncertainties in separating natural and anthropogenic contributions to biomass burning over the past 150 years (Dorothy Koch, personal communication). Thus, we perform similar simulations to Exp CC2 but with 10% of biomass for pre-industrial simulations. As compared with $\Delta\text{Exp CC2}$, the AIE is now slightly negative (-0.07 W/m^2).

Nober *et al.* (2003) find no significant signal in the radiation budgets in their simulations, although it is not clear if they refer to changes between simulations with and without aerosol-convective cloud effects or changes with respect to

changing aerosol concentrations. In our case, we do find increased cloud radiative forcings (more negative) for Exp CC1 and CC2 compared with Exp A, and smaller cloud radiative forcings for $\Delta\text{Exp CC1}$ and $\Delta\text{Exp CC2}$ compared with $\Delta\text{Exp A}$. Although aerosol-convective cloud effects may appear to be radiatively unimportant in their simulations, Nober *et al.* (2003) do find a weakening of the Walker circulation during June to August caused by the negative precipitation anomaly of the summer monsoon over the Indian subcontinent and surrounding regions including the southwest Pacific, which are correlated to changes in the velocity potential fields. In our simulations, zonally averaged profiles of vertical velocity and vertical transport of latent heat for June to August indicate increased vertical velocities and increased vertical transport of latent heat in the Southern Tropics for Exp CC1/CC2 versus Exp A, along with increased water contents at higher atmospheric levels. Comparison of observed June–July–August precipitation fields (e.g. from the global precipitation climatology project, http://precip.gsfc.nasa.gov/gifs/sg_v2.0259.gif) with present-day simulations for Exp A, CC1 and CC2 (see Figure 3.3a in colour plate section) indicates that Exp CC2 may exaggerate the reduction in precipitation in the Amazonia, but Exp CC1 does simulate the precipitation amounts over the tropical convective regions and mid-eastern Amazonia slightly more realistically than Exp A. This can be seen in Figure 3.3b in the colour plate section, which shows the difference in precipitation between Exp CC1 and Exp A (the decrease and increase in precipitation over the biomass burning regions are in the same direction as observations).

3.3.3 Regional impacts of aerosols on clouds and climate

Changes in global annual values may obscure significant regional as well as seasonal changes. Thus, it becomes necessary to focus on regional changes within a global model since although aerosol sources vary regionally, transport and teleconnections may play a role in modifying climate at a distance. Such changes have been found in several other studies that investigated the role of Asian and European/Russian aerosols in affecting the distant Arctic climate (Koch and Hansen, 2005). Increases in aerosol concentrations in North America via transport of dust plumes (that include entrained pollutants) from Mongolia and China (DeBell *et al.*, 2004) have also been observed. Seasonality can also play a strong role in the types of aerosols produced (fossil or biomass based production) and thus may change values of the resulting radiative fluxes (Menon, 2004). Here, we focus on those regions that have the largest changes in either biomass or fossil- and biofuel based aerosols. The seasons are chosen based on the availability of observations from large field campaigns so that model simulations of certain variables may be compared to observed values.

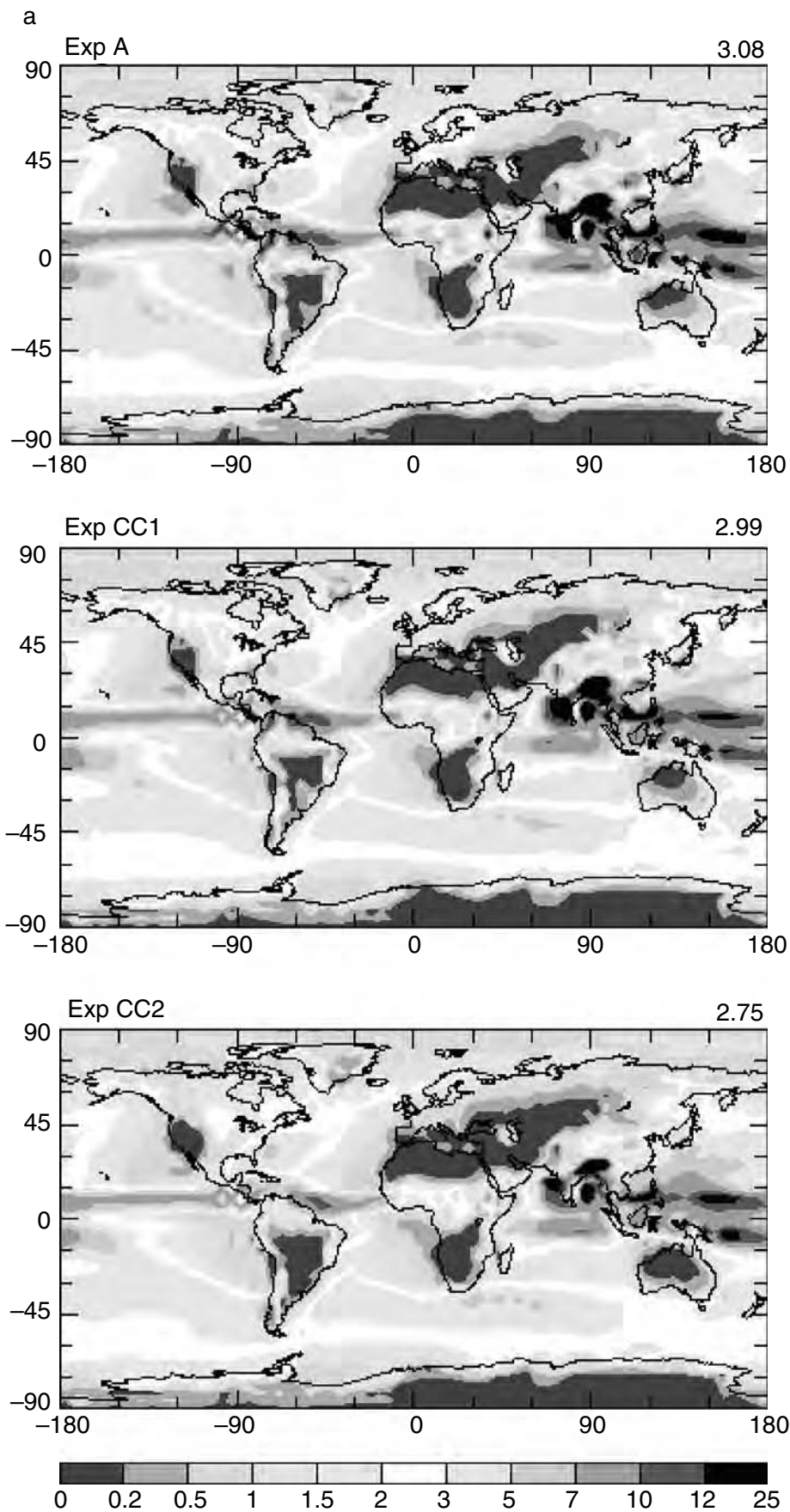


Figure 3.3 June–July–August precipitation (mm/day) fields for the year 2000 from Exp A, Exp CC1 and Exp CC2 (a), and change in precipitation between Exp CC1 and Exp A (b). Global mean values are indicated on the right-hand side.

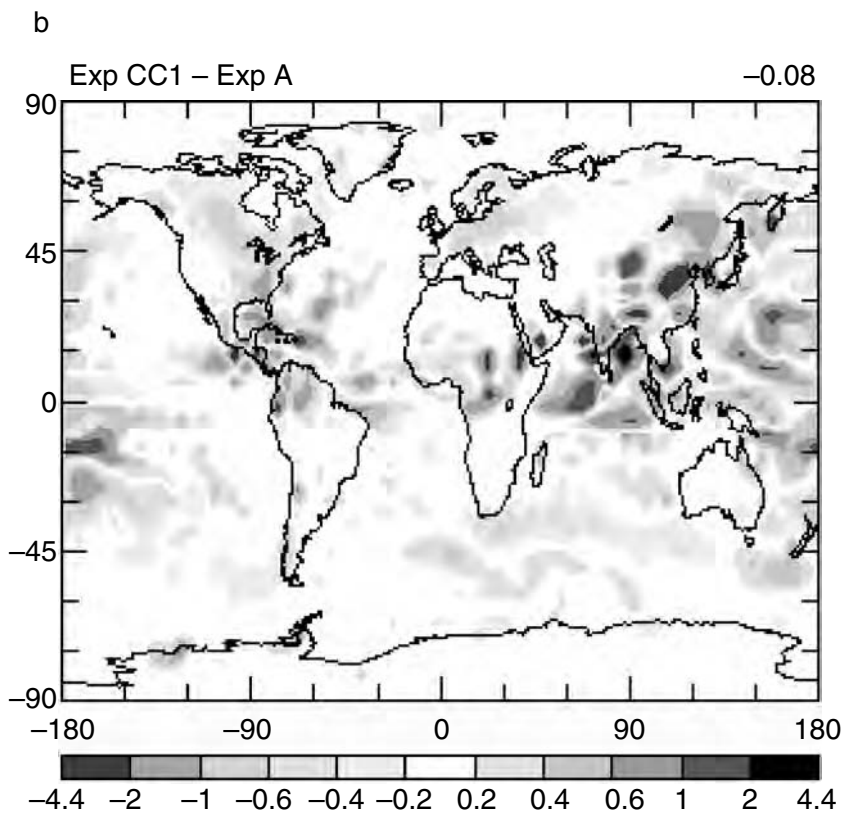


Figure 3.3 Continued

Black carbon aerosol effects on regional climate

Aerosol effects over China are thought to be partly responsible for the anomalous changes in the precipitation field observed over the past 30 years, with the north being more prone to droughts, while the south is subject to large-scale flooding (Xu, 2001; Menon *et al.*, 2002b). Several other studies have investigated changes in sunshine duration, diurnal cycle of temperature, and the anomalies of the northwestern Pacific subtropical high (Kaiser and Qian, 2002; Gong *et al.*, 2004), which are correlated to these anomalous precipitation fields. Similarly, over the Indian subcontinent and surrounding regions, Ramanathan *et al.* (2001) and Chung *et al.* (2002) find large spatial changes in the distribution of precipitation related to changes in surface forcing, from absorbing aerosols.

Here, we show values of radiative fluxes at the surface, TOA, and the atmospheric column along with precipitation for select regions over India and China in Tables 3.4a and b, respectively. The region over China was based on prior results from Menon *et al.* (2002b) and Menon (2004) to highlight changes in the northern/southern regions based on changes due to BC. Over India, we chose a region based on field data that were available from the Indian Ocean Experiment (INDOEX) as given in Ramanathan *et al.* (2001). During January to March 1999, radiative forcings at the surface, TOA and atmosphere were -20 , -2 , 18 W/m^2 for the direct aerosol effect and -6 , -5 , 1 W/m^2 when including the first aerosol indirect effect

Table 3.4a Simulated changes in the top of the atmosphere (TOA), surface, and atmospheric net radiation budgets (W/m^2), and precipitation over India/Indian Ocean region.

Indian Ocean				
(Jan–Mar) 0–20° N, 40–100° E	ΔTOA (W/m^2)	$\Delta\text{Sfc.}$ (W/m^2)	$\Delta\text{Atmos.}$ (W/m^2)	$\Delta\text{Prec.}$ (mm/d)
Exp A	–2.97	–7.33	4.36	0.35
Exp NBC	–2.07	–3.52	1.45	–0.08
Exp 2BC	–2.06	–5.71	3.65	0.01

Table 3.4b Similar to Table 3.4a but for differences over northern and southern China.

China (Jun–Aug) 90–120° E		ΔTOA (W/m ²)	$\Delta\text{Sfc.}$ (W/m ²)	$\Delta\text{Atmos.}$ (W/m ²)	$\Delta\text{Prec.}$ (mm/d)
Exp A	34–42° N	−6.46	−5.38	−1.08	−0.39
	18–30° N	−5.69	−7.77	2.08	0.05
Exp NBC	34–42° N	−7.81	−5.17	−2.64	−0.10
	18–30° N	−6.56	−6.90	0.34	0.24
Exp 2BC	34–42° N	−3.87	−6.48	2.61	−0.77
	18–30° N	−5.41	−9.28	3.87	−0.13

(Ramanathan *et al.*, 2001). Values as high as 33 W/m² have been reported based on observed radiative fluxes and aerosol properties at a site in India during the dry season (January–April) (Pandithurai *et al.*, 2004) for the aerosol direct effect. In our simulations, the forcings computed for the Indian sub-continent and Ocean region for the January–March period, shown in Table 3.4a, are for differences between present-day and pre-industrial aerosol emissions, similar to those from Ramanathan *et al.* (2001). As shown in Table 3.4a, our TOA and atmospheric fluxes are generally higher than those obtained from INDOEX. In both regions, precipitation appears to increase with increasing atmospheric fluxes (reduced surface fluxes), which may be partly due to the response of convection to the atmospheric heating, although it is not that easily demonstrated without separating it from the feedbacks that exist when linking the dynamical response of convection to heating or stability conditions, as shown in Chung *et al.* (2002). Over China, precipitation decreases more in the north than in the south, consistent with changes in the atmospheric fluxes for each of the simulations, although there is no direct relationship between them across the north and south domain. This suggests that along with the atmospheric aerosol content, surface and meteorological conditions also play an important role in modifying the response of precipitation to atmospheric heating.

Other climate aspects of absorbing aerosols are the direct heating effects on cloud cover – also called the “semi-direct effect” (Hansen *et al.*, 1997). Although the semi-direct effect (calculated as the difference between TOA net irradiance for simulations with and without aerosols, due to changes in cloud fields) (Johnson *et al.*, 2004) may be associated with positive forcing (due to changes in cloud cover that can increase the absorbed shortwave), recent studies have suggested that the sign may be dependent on the vertical aerosol distribution. In Tables 3.5a and 3.5b we show results

Table 3.5a Local radiative forcings for different vertical aerosol distributions.

Condition	Semi-direct effect (W/m ²)	Direct effect (W/m ²)
Absorbing aerosol in boundary layer	22.8	20.5
Absorbing aerosol in and above boundary layer	10.2	15.4
Absorbing aerosol above boundary layer	−9.5	0.7
Scattering aerosol above boundary layer	−0.1	−7.4

Source: Johnson *et al.* (2004)

Table 3.5b Local radiative forcings for different vertical distribution of aerosols over China (18–50° N, 90–130° E).

Condition	Semi-direct effect (W/m ²)	Direct effect (W/m ²)
Standard aerosol distribution	−5.39	5.82
All aerosols in first layer (0.42 km)	0.97	−0.06
All aerosols in fourth layer (3.8 km)	−5.74	8.37
All aerosols in seventh layer (10.5 km)	−19.8	15.3

Source: Menon, (2004).

from a large eddy scale model (Johnson *et al.*, 2004), and a GCM based study (Menon, 2004) that investigated regional forcings associated with different vertical distributions of absorbing aerosols. As shown in Table 3.5a, with absorbing aerosols above the boundary layer, the semi-direct effect can be negative, mainly owing to the increases in liquid water path (LWP) fields and entrainment effects that led to a shallower moister boundary layer with higher LWP (Johnson *et al.*, 2004). In the GCM study, the effects of vertical velocity, which got stronger as aerosols were distributed higher, led to increased cloud formation and LWP, which in turn changed the semi-direct effect from positive to negative (Menon, 2004). Penner *et al.* (2003) also find negative semi-direct effects for biomass aerosols injected in the mid troposphere, mainly by accounting for the longwave forcing (longwave forcing decreases when high cloud cover decreases).

On a global average, the semi-direct effect, as indicated in a simulation comparable to Exp A but without the indirect aerosol effects (Exp NIE-CDNC is kept a constant 174 cm^{−3} for land and ~60 cm^{−3} for ocean), is approximately −0.08 W/m². Regions of negative values are co-located with high biomass burning regions, and changes to both longwave cloud radiative forcing and high cloud cover are negative compared to ΔExp A. Thus, these negative longwave effects outweigh the positive forcings due to cloud reduction, which then give us a semi-direct effect that is negative in regions of strong biomass burning. However, Koren *et al.* (2004) find that for heavy smoke conditions in the Amazonia, the cumulus cloud cover decreases by 38% as compared to clean conditions, with an overall instantaneous forcing of 8 W/m². Another study in the Middle East during the Kuwait oil fires found that depending on the stability of the atmosphere and humidity profiles, large deep convective clouds can form at the top of the smoke layer (Rudich *et al.*, 2003). Thus, vertical profiles of temperature

(heating), (and thus of aerosols) and humidity may play a leading role in changing the sign of the semi-direct aerosol effects due to changes in cloud cover. These various results are difficult to interpret since observations of vertical aerosol distributions remain sparse.

Effects of biomass aerosols over Amazonia

During the active fire season large amounts of forest are burned in Amazonia, and changes in carbon fluxes, as well as cloud properties, are significantly different from the areas not affected by burning as shown by satellite records (Kaufman and

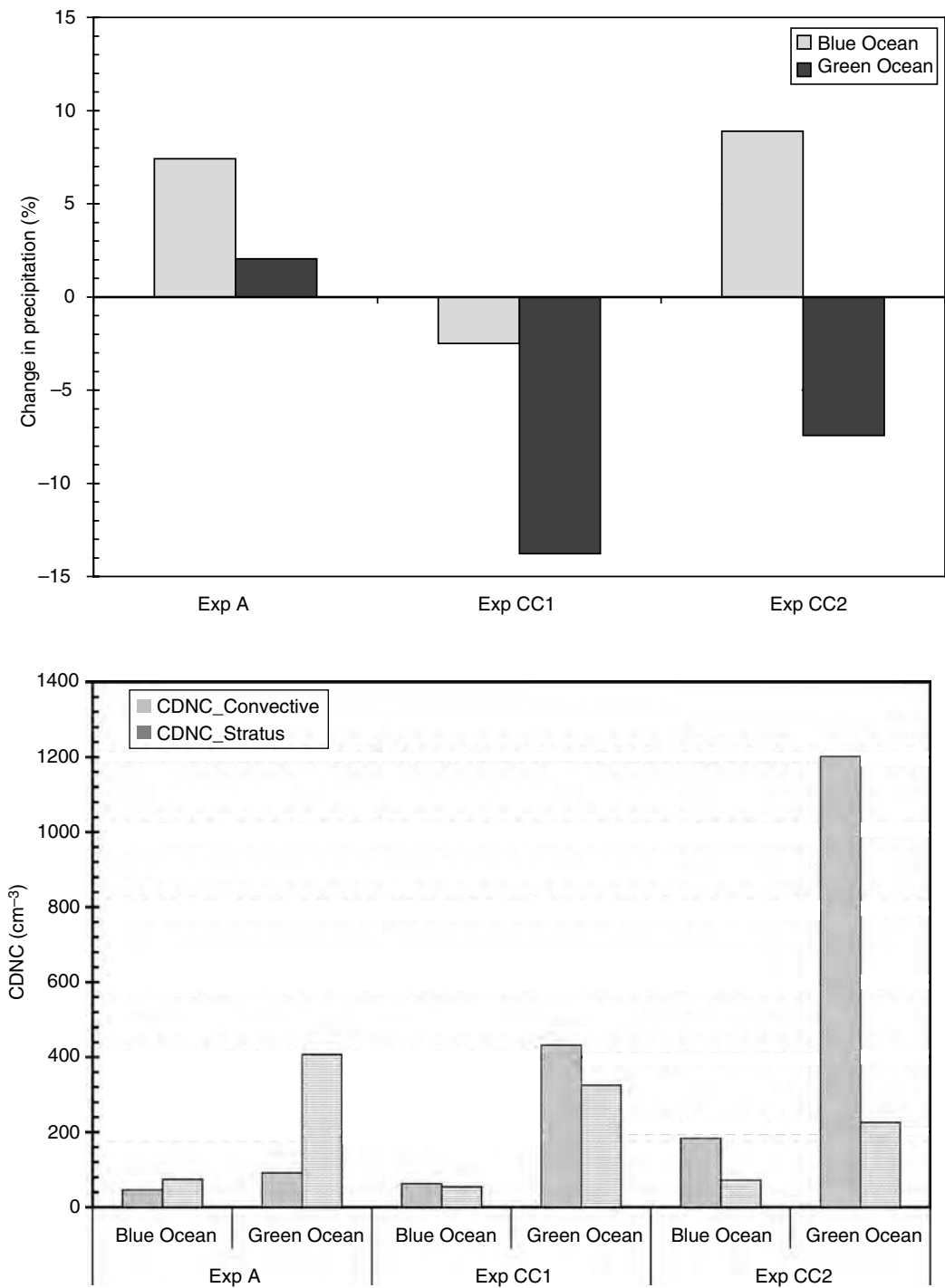


Figure 3.4 Change in precipitation and cloud droplet number concentration (CDNC) for the ocean (Blue Ocean) and the biomass source (Green Ocean) regions for the different model simulations.

Fraser, 1997). Recent results from a large field campaign conducted over Amazonia during the fire season indicate changes in deep convective cloud properties and precipitation when Amazonia was separated into a green ocean (biomass source region), a blue ocean (ocean) and a polluted smoky region (Andreae *et al.*, 2004). Separating Amazonia similarly, on the basis of aerosol concentrations, we compare model computations of precipitation and CDNC for simulations with and without aerosol convective cloud effects in Figure 3.4. Since we use seasonal means for August to October, specific days of active burning are not accounted for and thus, we mainly contrast values for the ocean (5°N – 5°S , 30 – 40°W), “blue ocean”, versus regions with large biomass emissions (5 – 15°S , 50 – 75°W), “green ocean”. We compare our CDNC values with those from the SMOCC campaign (Andreae *et al.*, 2004) of 200, 600 and 2200 cm^{-3} for the blue ocean, green ocean and smoky clouds, respectively. The CDNC values for Exp CC1 and CC2 are comparable to those measured within uncertainties that are bound to exist when comparing single point/time values with averages from the model over larger domains.

As shown in Figure 3.4, precipitation changes from the ocean to the biomass regions are stronger for Exp CC1 and CC2, mainly because of the substantial increase in CDNC obtained for convective clouds. Without aerosol–convective cloud effects, no decrease in precipitation is evident. In addition, zonally averaged vertical profiles of particle size indicate that the size of particles needed to participate in coalescence processes (12 – $14\text{ }\mu\text{m}$) is at $\sim 900\text{ hPa}$ in Exp A, but between 600 and about 900 hPa in both Exp CC1 and CC2, indicative of a shift in onset of precipitation. Andreae *et al.* find a shift in the onset of precipitation under smoky conditions that tend to favor the formation of large ice hydrometeors, although the change in the amount or areal distribution of precipitation was not evident, because the increased intensity of storms compensates for the effects of the suppression of precipitation initiation (time and level for cloud droplet sizes to reach the coalescence size). As shown in Figure 3.4, both $\Delta\text{Exp CC1}$ and CC2 indicate a larger decrease in precipitation from the ocean region to the biomass regions compared with that seen in $\Delta\text{Exp A}$. However, verification of such changes with observations is not possible and thus these values are mainly indicative of the sensitivity of precipitation to changing CDNC in convective clouds.

3.4 Conclusion

In the newly updated version of the GISS GCM, the aerosol indirect effect was found to be -0.65 W/m^2 (or alternatively -1.36 W/m^2), much lower than previous values obtained in Menon *et al.* (2002a) that ranged from -2.4 to -4.4 W/m^2 . These lower values were mainly due to smaller differences in the anthropogenic aerosol burdens currently assumed, changes in the aerosol extinction efficiency due to aerosol effects on relative humidity, (36%), and addition of BC and dust aerosols, amongst other general improvements in the updated

model. Including aerosol effects on convective clouds shifts the cloud water content to higher levels, as well as changing the initiation of precipitation formation to higher levels. The percentage change in precipitation for the clean ocean regions versus the biomass regions was found to vary by a factor of 2 to 4 mainly owing to the increase in cloud droplets in convective clouds. The aerosol indirect effect with aerosol–convective cloud effects included ranged from $+0.22$ to -0.43 W/m^2 and was much lower than estimates without aerosol–convective cloud effects, some of which may be related to changes in the longwave cloud forcing, which was of similar magnitude to the shortwave cloud forcing. The global magnitude of the aerosol semi-direct effect was found to be close to 0 (-0.08 W/m^2) since longwave effects of biomass aerosols (lofted higher) cancel out positive forcings from a reduction in cloud cover. Changes in the vertical profiles of temperature and humidity as well as the liquid water path can influence the sign of the semi-direct effect through changes in the cloud cover.

Several simulations were conducted to examine the effects of BC aerosols from fossil- and biofuel and carbonaceous aerosols from biomass burning. For twice the amount of fossil- and biofuel BC emissions, with all indirect effects included, the ratio of surface temperature to TOA radiative fluxes (for present-day versus pre-industrial aerosol emissions) increased by more than a factor of 10 (changes from 0.097 to $1.14\text{ K W}^{-1}\text{ m}^2$). While the forcing associated with BC may be somewhat linear, the temperature response to the forcing is non-linear since it includes changes in cloud fields and other feedbacks. Climate sensitivity to doubled CO_2 in a similar version of the model coupled to a slab-ocean model is $\sim 0.66\text{ K W}^{-1}\text{ m}^2$ (Hansen *et al.*, 2005). These results and those from Roberts and Jones (2004) (climate sensitivity of 0.56 and $0.91\text{ K/(W/m}^2)$ for four times the fossil fuel BC amount and twice the CO_2), as well as results based on the forcing efficiency of BC (Table 3.3) and Jacobson (2002), indicate that despite small atmospheric burdens ($\sim 0.13\text{ mg/m}^2$) and short life times (~ 3 – 7 days), BC aerosol emissions have a significant influence on climate. Regional changes in precipitation, examined over India and China, were found to be related to the amount of atmospheric heating, with higher atmospheric fluxes corresponding to larger changes (positive) in precipitation, though we do not discount the influence of surface and meteorological conditions that may also lead to similar changes. The changes described in this paper are for aerosol effects on warm clouds only, and to fully understand the net aerosol effects on climate, various aerosol effects on cold cloud microphysics must be included in future simulations.

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